

# Year 9 Science – Booklet 12: Biology

## Cell Biology — ANSWERS

### Quick Test (p.5)

1. Any three of: plant cells have a cell wall, a (large) vacuole, and chloroplasts, which animal cells don't have.
2. It holds the cell together and controls what passes in and out of the cell.
3. It gives the plant cell strength and support (made of cellulose).
4. A cell that has changed shape/structure to carry out a particular job efficiently.
5. Tissue.
6. Mitosis.

### Section A – Multiple Choice (5 marks)

1. a) cell membrane, nucleus
2. c) controlling what passes in and out of the cell
3. b) tissue
4. b) to absorb sunlight for photosynthesis
5. d) biconcave

### Section B (15 marks)

#### 1. Missing labels on plant cell diagram (3 marks)

A: **Cell wall** (outer boundary). B: **Chloroplast** (small oval structures in the cytoplasm). C: **Vacuole** (large central space).

(Check these against your own diagram – letter positions can vary between printings.)

#### 2. Which cells contain which parts? (12 marks)

|                 | Chloroplasts | Nucleus | Cell wall | Mitochondria |
|-----------------|--------------|---------|-----------|--------------|
| Onion cell      | ✗            | ✓       | ✓         | ✓            |
| Palisade cell   | ✓            | ✓       | ✓         | ✓            |
| Human skin cell | ✗            | ✓       | ✗         | ✓            |

Onion cells are plant cells but don't photosynthesise, so no chloroplasts. All plant cells (onion, palisade) have a cell wall; animal cells (human skin) don't. All three have a nucleus and mitochondria.

### Section C (12 marks)

#### 1. Red blood cell

- a). A nucleus (2 marks) – red blood cells lack one, unlike other animal cells, to leave more room to carry oxygen-binding haemoglobin.
- b). To carry oxygen around the body.
- c). The cell wall gives the plant cell strength and support, and helps maintain its shape.
- d). Chloroplasts absorb sunlight so the plant can carry out photosynthesis and make its own food.
- e). Any two of: root hair cells, palisade cells.

#### 2. Three other specialised animal cells (3 marks)

Any three of: sperm cells, nerve cells, egg cells (ova), muscle cells, white blood cells.

## B1.1 Observing cells

### *In-text questions*

- A. Cells.
- B. Tiny roomlike structures, which he called “cells” (the empty box-like cell walls of the dead cork tissue).
- C. Looking carefully and in detail at an object (and recording what is seen).
- D. The eyepiece.

**Magnification box: eyepiece ×10, objective ×50**

Total magnification =  $10 \times 50 = 500$ .

### *Summary Questions*

#### **1. Copy and complete (5 marks)**

All living organisms are made up of **cells** – these are the building blocks of life. To **see** cells in detail you need to use a **microscope**. This **magnifies** the object.

#### **2. Functions of microscope parts**

- a). Lenses: magnify (enlarge) the image of the object.
- b). Stage: the flat platform that holds/supports the slide and specimen.
- c). Focusing knobs: move the stage/lens to bring the object into focus – coarse focus for a rough adjustment, fine focus for a precise one.

#### **3. Method to observe cells in a white flower petal (6 marks QWC)**

Cut or peel a very thin section of petal tissue (thin enough for light to pass through); place it flat on a slide; add a drop of water (and a stain if needed to make the cells more visible) and lower a cover slip on at an angle to avoid air bubbles; place the slide on the stage and secure with the stage clips; select the lowest-power objective lens first; look through the eyepiece and turn the coarse focus knob until the cells come roughly into view; use the fine focus knob to sharpen the image; then repeat with a higher-power objective lens for more detail, refocusing with the fine focus knob each time.

## B1.1 workbook

### **A. Fill in the gaps**

All living organisms are made up of **cells** – these are the building blocks of life. To see cells in detail you need to use a **microscope**. This **magnifies** the object. Looking carefully and in detail at an object is called making an **observation**.

### **B. Match labels W, X, Y, Z**

slide → **Y** • eyepiece lens → **W** • objective lens → **X** • light → **Z**

(Check against your own diagram – based on W at the top of the tube, X at the rotating lens turret, Y at the stage, Z at the base.)

### **C. Henry observing a leaf**

- a). Select the objective lens with the lowest magnification, then look through the eyepiece and slowly turn the coarse focus knob until the leaf comes into view.
- b). Switch to a higher-magnification objective lens, then use the fine focus knob to sharpen the image.
- c). Total magnification =  $10 \times 50 = 500$ .